Evelyn Du

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Operational Research and Industry Engineering, **GPA: 4.0** **Jan. 2026**
Renmin University of China, Beijing, China
Bachelor of Science in Accounting & Data Science, **GPA: 3.8** **June 2024**

Universitidal Isabel, Master of Big Data and Intelligence, **GPA: 9.93/10** **June 2024**
University of Maine at Presque Isle, Bachelor of Arts in Management Information Systems, **GPA: 3.93** **June 2024**

Selected Coursework: Machine Learning for Big Messy Date, Reinforcement Data Mining, Computational Optimization, Risk Modeling, ETL Process, Blockchain Practice, Business Intelligence Simulation, Robot Automation Design

Selected Publication:

- Bitcoin Arbitrage: The Role of a Single Exchange: *Link, In 2026 Southwest Finance Association Conference*,March 2026
- Application of three social psychological models in post-epidemic, based on statistical science summary test. *Journal of Humanities and Social Sciences,* September 2024
- Tokyo Stock Exchange Prediction with Hybrid Models of Lightgbm and DNN, *the 2nd International Conference on Mathematical Statistics and Economic Analysis,* May 2023.
- Application of support vector regression in prediction model using a genetic algorithm optimized, *Journal of Physics: Conference Series,* July 2021
- Economic Forecast of the Southern China on BP Neural Network---Taking Chongqing as an Example,In *6th International conference on financial innovation and economic development,* March,2021
- A new image description generation algorithm based on improved attention mechanism, *In Basic & Clinical Pharmacology & Toxicology,* January 2021

Patent:

- Du, W., Chu, Q., Qin, Y., Li, X., Guan, Y., Chen, H., & Xia, J. (2022). A novel multifunctional smart bracelet [Patent No. ZL 2022 2 0604426.8]. Patent application date: March 18, 2022.
- Chu, Q., Du, W., Wei, T., Jiao, Y., & Li, M. (2021). A heating load forecasting system based on the LSTM algorithm model V1.0 [Software]. Development completed: July 15, 2021; First published: July 16, 2021.

Research Interests:

- Machine Learning for Decision Making in Finance, Computational Optimization for Supply Chain, Complicated Algorithm

Awards:

- Akuna Trading Competition Silver (2024), Association for Computing Machinery Member (2023), Mathematical Contest in Model F Prize (2022), Global Citizen Scholar (2021), Kaggle (Silver, 2021), Mathematical Competition (1st prize, 2018)

SKILLS & INTERESTS

- **Programming Languages and Software:** Python (Pandas, NumPy, Matplotlib, TensorFlow, PyTorch; SVD, KNN), R (Regression Analysis, Decision Tree), SQL, Git, STATA, SPSS, MATLAB; Jupyter Notebook, Bloomberg, Wind, PointBI
- **Data Visualization:** Tableau, Power BI, Web (Jekyll, Hugo, Hexo, WordPress, Vue), Figma, HTML, JavaScript
- **Languages:** Mandarin (native), English (fluent), Korean (intermediate), French (basic)
- **Financial Certification:** ACCA(AB/MA/TA/LW/PM/TX/FR); CIMA; CPA Candidate
- **Interests:** Marathon Running; Scientific Novel Writing; Travelling

RESEARCH EXPERIENCE

AI-Powered Financial Research Assistant, *Reportify* **Jan 2025 – Present**
Supervisor: Dr. Liang Xu — Founder of FinScope Labs, Harvard University (Statistics)
Mentee of Nobel laureate in Economics

- Built a multi-step fact-checking financial agent capable of three-pass web search, cross-document verification, and structured evidence extraction. It automatically validated figures in earnings releases and investor presentations, reducing analyst verification time by ~60%.
- Developed an automated corporate-filings analysis agent, combining Python, regex parsers, and LLM reasoning to process 600+ U.S. 10-K/10-Q filings. The system identified revenue trends, margins, and risk disclosures, generating concise summaries and cutting report-processing time by ~70%.
- Integrated cross-modal retrieval + reasoning, aligning text extraction, basic financial modeling, and sentiment analysis. Earnings-call transcripts were embedded into a vector database; the agent produced sentiment scores, linked them to price reactions, and auto-generated plots—mitigating common multi-tool coordination issues noted in recent agent research.
- Conducted causal-inference research on crypto-equity spillovers, assembling 7,500+ crypto and 6,000+ U.S. equity/ETF datasets (2015–2025). Using Double Machine Learning, the agent built rolling panels and ran statistical tests autonomously. Results showed post-ETF-launch contagion coefficients dropped >80%, indicating ETF-mediated risk absorption.
- Shipped a user-facing dashboard via Streamlit and Reportify's Agent Builder enabling instant financial summaries from custom company lists. Collected feedback from 20+ industry users, iterated on prompt logic and entity-matching rules, improving satisfaction from 3.8 → 4.5.
- Implemented safeguards to reduce hallucinations, including cross-filing validation and confidence-based flagging. Given that leading agents still achieve only ~48.9% accuracy on complex financial tasks and Reportify's weighted-average benchmark score is ~27.7% vs. 84.7% for human experts, adding verification loops significantly improved reliability.

Research Assistant — AI for Healthcare *(MedX-PKU × University of Edinburgh CMI)* **Aug 2025 – Present**
Advisors: Dr. Liantao Ma (PKU MedX) & Dr. Junyi Gao (UoE CMI)

- **Goal:** Develop trustworthy clinical AI by integrating ICU waveforms + EHR time-series for interpretable risk prediction and early warning.
- **Benchmarking:** Constructed a waveform–EHR benchmark of 12 000 patients / 58 000 h of high-freq signals (125 Hz); standardized cohorting, windowing, and dual evaluation (AUROC/AUPRC + MSE/CC). Reduced data-prep time by 72 %.

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- **Event Prediction:** Built LSTM and Transformer models for hypoxemia and hypotension, achieving AUROC 0.87 (+6 % vs baseline) and AUPRC 0.61 (+9 %) on MIMIC-IV; validated cross-hospital transfer on VitalDB with ΔAUROC +0.04.
- **Autoregressive & SSL Modeling:** Trained masked-signal models on ECG/PPG/ABP/RESP; improved downstream F1 by 14 % under 30 % label fraction and cut missing-data error by 25 %.
- **Trustworthiness:** Applied calibration (ECE ↓ 38 %), uncertainty ensembles, and clinician-readable explanations, reducing false-alert rate by 22 % at 90 % sensitivity.
- **Agentic LLM Pipeline:** Deployed multi-agent EHR analytics workflow with function-calling and self-verification, shrinking analysis cycle from 3 days to 5 hours.
- Open Tools: Released benchmark package and PyHealth/RHealth extensions (> 150 GitHub stars, 10+ institutional users).

Tech: Python (PyTorch, Lightning, WFDB, Spark, Hugging Face), RHealth, SQL, Docker, MLflow. Keywords: ICU waveforms, EHR, benchmarking, self-supervised learning, LLM workflow, trustworthy AI.

Research Assistant; *LLM "Coding Agent" – Multi-Modal Reasoning and Reinforcement Lab* **Aug. 2025 – Present**
Advisor: Dr. Zeyu Zhang (Applied Scientist, Amazon AGI; Ph.D., University of Arizona) & Dr. Akide Liu (Ph.D. Candidate, Monash University; Efficient AI & Computer Vision Lab)

- **Research Goal:** Investigate how multi-modal data (code + GUI + language) enhances reasoning, verification, and self-correction in large language models (LLMs).
- **Benchmarking & Model Development:** Evaluated SWE-Agent/Bench, QSWE-QA across 11 000 + GitHub repositories; built cross-file dependency benchmark (∼5 M code relations). Integrated Curriculum RL and Self-Verification Loops (syntax, compilation, logic) — cutting post-execution failure rate from 38 % → 14 % and boosting pass@1 by +12 %.
- **Diffusion LLMs & Trajectory Engineering:** Implemented hybrid diffusion–autoregressive architectures (LLaDA-MoE, RLVR) to generate structured multi-turn reasoning traces. Achieved +9 % task-completion, −28 % reward-variance, and +16 % context-recall on 3 RL datasets (∼60 K episodes).
- **Reflection Framework Building:** Designed Reflexion + GUI-Reflection pipeline for adaptive code debugging; enabled self-feedback every 3 iterations, reducing human-in-loop edits by 45 %.
- **Multi-Agent Reinforcement Research:** Co-developed "LLM 翰林院" (LLM Hanlin Academy) simulation platform; released > 40 agents for code, planning, and GUI tasks; community adoption > 2 K downloads, GitHub 300 +.
- **Impact:** Produced 2 internal technical reports and 1 under-review NeurIPS Workshop paper; deployed experimental agent workflow used by 4 collaborating labs (Amazon AGI, Monash AI4Code, PKU MedX, UIUC AI4SE).

Tech: Python (PyTorch, LangChain, Ray RLlib), GitHub Actions, Docker, WandB, SQL, Gradio, FastAPI.
Keywords: multi-modal reasoning, LLM agents, Curriculum RL, diffusion modeling, self-verification, code reasoning

Research Assistant; *National University of Singapore* **Nov 2024 – May 2025**
Advisor: Dr. Hanzhang Qin (Assistant Professor of Industrial Systems & Engineering; Ph.D., MIT Computational Science & Engineering) – *Digital Trust Center (GenAI + Blockchain Security)*

- **Blockchain Security & Smart Contract Benchmarking**
  - Developed a large-scale benchmark dataset for LLM-based smart contract vulnerability detection, integrating 1.2M on-chain transactions (Ethereum & BNB Chain), 250K+ wallet addresses, and 3,500+ real-world smart-contract scripts and whitepapers.
  - Curated and released ChainTrust-Bench (220K labeled transactions) for training and evaluation of security-focused LLM agents across 5 benchmark datasets
- **LLM-Driven Multi-Agent Security Systems**
  - Designed and implemented LLM-based security agents (GPT-4) acting as automated Defender Agents for smart contract auditing, policy-based risk classification, and access-control simulation.
  - Conducted agent-driven vulnerability analysis covering reentrancy, front-running, oracle manipulation, and unauthorized access, achieving precision 0.93 and improving threat-detection recall by +18% over rule-based baselines.
  - Reduced manual audit workload by 65% through agent-based reasoning and automated security evaluation.
- **Empirical Validation & Simulation**
  - Evaluated LLM security agents through large-scale empirical validation on real-world blockchain data, modeling common attack archetypes via causal inference and anomaly detection, achieving AUC = 0.92.
  - Supported deployment-oriented security screening pipelines for decentralized exchanges and token-launch platforms, bridging offline benchmarks with realistic on-chain security scenarios.
- **Empirical Data Analytics**
  - Evaluated LLM security agents through large-scale empirical validation on real-world blockchain data, modeling common attack archetypes via causal inference and anomaly detection, achieving AUC = 0.92.
  - Supported deployment-oriented security screening pipelines for decentralized exchanges and token-launch platforms, bridging offline benchmarks with realistic on-chain security scenarios.
- **Research Output & Impact**
  - Co-authored 1 working paper (LLM Agents for On-Chain Security Reasoning, under review, IEEE S&P 2025).
  - Open-sourced ChainTrust-Bench, adopted by 2 fintech startups for live contract screening and risk assessment.

Tech: Python, Web3.py, Etherscan API, Solidity, Node.js, Hugging Face, LangChain, Neo4j, SQL, PyTorch, scikit-learn.
Keywords: blockchain security, smart-contract auditing, anomaly detection, GenAI for Trust, financial risk modeling.

Research Assistant; *Michigan State University- Eli Broad College of Business* **Aug. 2025 – Present**
Advisor: Dr. Siqi Pei (Assistant Professor of Marketing; Ph.D., CUHK Business School; Visiting Faculty Scholar, MIT Sloan)

Project 1 – Large Language Model Suggestions and Human–AI Interaction in Sales Communication

- Designed and executed a field-experiment framework examining how AI-generated suggestions affect sales consultants' communication styles and customer-conversion outcomes across >3,000 recorded sales interactions.
- Constructed a taxonomy of AI suggestion labels capturing not only linguistic tone but also informational framing (e.g., benefit-focused vs. limitation/negative framing), using transformer-based embedding models to classify LLM-generated messages (RoBERTa / GPT-4 embeddings + K-means clustering, F1 = 0.87).
- Integrated causal-inference methods (DiD / A/B testing) to quantify how different suggestion styles influenced downstream sales metrics (conversion +15 bps ↑, response latency ↓12%).

- Mechanism analysis shows that suggestion styles emphasizing constraints, trade-offs, or potential downsides improve performance by reducing customers' cognitive load and accelerating product-fit screening, rather than by increasing persuasion intensity.
- Conducted an extensive literature review on human–AI collaboration and assistance, synthesizing 60+ UTD24/FT50 articles to support theoretical framing around informational usefulness, interpretability, and human decision authority.

Project 2 – Cross-Channel Synergy in Digital Automotive Marketing

- Integrated multi-platform digital marketing datasets (>5M ad impressions) across broad-reach entertainment media (Douyin, Kuaishou, Weibo), diagnostic vertical platforms (Dongchedi, Autohome, Yiche), and brand-owned channels (Nissan App, WeChat mini-programs), capturing consumers' nonlinear and cross-channel automotive purchase journeys.
- Demonstrated that consumer search behavior exhibits cross-stage transitions—from attention-oriented platforms to diagnostic search engines to brand-owned commitment channels—contradicting traditional linear-funnel assumptions and highlighting dynamic, multi-touchpoint, and path-dependent decision patterns.
- Identified non-intuitive channel ROI dynamics, where vertical media—though displaying the lowest standalone conversion—generated strong positive spillover effects onto owned channels (interaction coefficient $\beta \approx 0.42$, $p < 0.01$), revealing their evaluation-stage catalytic function in durable-goods decision processes.
- Developed a panel-data and cross-channel interaction modeling framework to estimate substitution vs. complementarity among channel pairs. Results show that information-rich diagnostic platforms increase subsequent brand-app purchase likelihood by ~20%, confirming their role as evaluation accelerators within the dynamic consumer decision system.

Research Assistant; *HKUST Business School – Management Department* **Aug. 2025 – Present**

Advisor: Dr. Siyin Chen (Assistant Professor of Management; Ph.D., Rotman School of Management, University of Toronto; Ethnographer specializing in moral markets and the sociology of professions)

- Designed and coordinated a multi-province study on AI-driven medical consultations, managing large-scale survey-style datasets (500+ sessions) and standardizing coding protocols across RA teams.
- Applied large language models (LLMs) and reproducible NLP workflows to analyze 100,000+ dialogue turns, identifying empathy cues and relational strategies with 85%+ classification accuracy.
- Integrated statistical hypothesis testing, causal inference methods, and qualitative coding, linking patient engagement strategies to 15–20% higher participation rates.
- Built replicable, automated NLP pipelines for multilingual transcripts (Mandarin/Cantonese/English), >90% accuracy and reducing manual labor by 70%, with infrastructure transferable to digitized surveys, census data, and social records.

Coauthor & Research Assistant*, Cornell University* **Jun. 2025 – Present**

Advisor: Irene Aldridge (M.S. Financial Engineering Professor, Columbia University; Author of High-Frequency Trading, Big Data Science in Finance; CEO of AbleMarkets Inc)

- Developed a risk-sharing contagion framework integrating Kou (2002) jump-diffusion and Chernozhukov et al. (2018) Double Machine Learning (DML) to quantify crypto-to-equity spillovers.
- Constructed 7,500+ cryptocurrency and 6,000+ U.S. equity/ETF datasets (2015–2025 period) using Yahoo Finance API; engineered 14-day rolling mean-variance panels and jump-return estimators under Itô-lemma-based risk adjustments.
- Implemented DML-based causal inference pipelines, isolating directional dependencies between BTC-USD, ETH-USD, and S&P 500 ETF (SPY), achieving reproducible estimations with >95% significance consistency across sub-samples.
- Identified structural regime shifts after Bitcoin and Ethereum ETF launches: contagion coefficients between crypto and equities declined >80%, demonstrating ETF-mediated risk absorption and reduced systemic fragility.
- Contributed to theoretical extensions of Kyle (1985) informed-trading model, proving equilibrium propositions on multi-asset liquidation and synthetic hedging through noise-trader liquidity dominance.

Teaching Assistant*, NBA6550 Business Data Analysis with SQL* **Aug. to Dec. 2024**

- Mentored 60+ students in SQL query design, data cleaning, and relational database management, ensuring consistent progress across diverse technical backgrounds.
- Created and distributed practice datasets modeled on census and historical surveys, strengthening student ability to manage and analyze large-scale structured social science data.

Research Assistant*; Stanford Institute for Economic Policy Research* **Jan. to Jul. 2024**

- Collaborated with Dr. Chenzi Xu to design and execute causal inference studies on 50,000+ World Bank transaction records spanning the 20th–21st century, coordinating workflows and replication protocols across 3 RA teams.
- Applied random forest, LSTM and econometric techniques (2SLS, DiD, hypothesis testing) to evaluate how international credit exporters reshaped trade flows during 5+ global financial crises, ensuring reproducibility and robustness of results.
- Synthesized empirical findings from multi-country panel data into 4 policy briefs and 2 internal seminars, translating statistical outcomes into accessible insights for both academic audiences and policy stakeholders, while contributing to cross-project intellectual exchange.

Research Assistant *,Cornell University* **Feb. 2022 to May. 2023**

- Assisted Dr. Yan Mi on education policy and Sino-American cultural exchange projects, coordinating multi-source data collection and coding protocols across 3+ RA teams to ensure standardized processing.
- Conducted quasi-experiments on local government education funding using clustering algorithms, 10,000+ comparison tests, and sigmoid sensitivity analysis, ensuring methodological rigor and reproducibility.
- Standardized analytical workflows for 50,000+ policy records, improving threshold efficiency by 20%, while fostering reproducible integration of statistical modeling with historical and education datasets.

Algorithm Project Leader*, Aqumon (the Top AI Trading Asian Company),* China **June to Dec. 2020**

- Led a project analyzing 10M+ financial transaction records, identifying 10 systematic trading risk points via SQL-based auditing and monitoring protocols that enhanced firm-wide risk oversight.
- Directed the training of decision-tree and econometric models on 5+ years of high-frequency financial datasets, achieving >50% yield in backtesting and presenting validated findings to 20+ institutional partners.

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- Established standardized data pipelines and documentation practices for high-frequency, unstructured datasets (100GB+), enabling reproducibility, smoother onboarding of new analysts, and scalability of empirical research workflows applicable to survey and policy datasets.

PROFESSIONAL EXPERIENCE

Quantitative Alpha Research Summer Intern, *Blackrock,* NewYork, the U.S. **June. to Aug. 2025**

- Designed a medium-term alpha strategy for structured products (MBS and fixed income packages) using proprietary alternative datasets, improving backtested Sharpe ratio by 0.25.
- Applied Bayesian regression, LSTM, and feedforward neural networks across 5+ market regimes (2018 – 2024), linking robustness tests to $10B+ AUM portfolios with systematic evaluation protocols.
- Prototyped a transformer-based LLM to optimize portfolio construction under asset-level constraints, achieving 12% volatility reduction and 18 bps risk-adjusted return improvement vs. benchmark.
- Coordinated with Modeling & Research and the Portfolio Analytics Group (PAG) to integrate LLM pipelines into Aladdin, developing documentation and workflows that enabled adoption by 3 portfolio management teams.

Investment Engineering Intern, *KKR (Kohlberg Kravis Roberts),* Shanghai, China **Jan. to May 2022**

- Analyzed 10M+ transaction records using a causal inference data governance model, identifying 8 purchasing personas that informed PE expansion strategy and presented findings through interactive Seaborn dashboards.
- Optimized ETL workflows in BigQuery with collaborative filtering techniques, raising model accuracy to $R^2 > 0.8$ and establishing standardized templates for future analysis.
- Partnered with investment analysts and data engineers to translate raw datasets into research-ready insights, building systems that supported reproducible applied econometric analysis for decision making.

Business Analysis Intern, *ByteDance,* Shanghai, China **June. to Sep. 2021**

- Facilitated cross-departmental data integration by implementing quality rules for procurement and sourcing systems, achieving 90%+ positive external feedback from stakeholders.
- Conducted A/B testing (SPSS) and applied Matlab moving-window models on K12 education datasets, standardizing experimental protocols to improve consistency across teams.
- Designed scalable Entity-Relationship Diagram frameworks to support analytics expansion, enabling tech teams to onboard new data pipelines and contributing to growth of the customer base to 30k+.

Quantitative Research Intern, *E Fund (the Largest Chinese Mutal Fund),* Guangdong, China **Feb. to June 2021**

- Automated large-scale financial data collection by creating a C++ scraping package, cleaning datasets from 1,000+ listed companies and 3,000 survey responses; standardized processes reduced manual workload across research teams.
- Built a Weighted Least Squares regression model from scratch to analyze P&L and evaluate cost control strategies for asset allocation, reducing commission fees by 10% and generating reproducible templates for future allocation studies.
- Developed scalable econometric modeling and automation pipelines for unstructured datasets, establishing workflows that mirrored census-style data processing and improved knowledge transfer between analysts.

Data Science Intern, *Boston Consulting Company,* Beijing, China **Jan. to May. 2020**

- Conducted industry, financial, and market analysis for a client feasibility study, synthesizing findings into a 52-page deliverable reviewed by senior consultants and external stakeholders.
- Processed multivariate tagged datasets with 20+ user-level indicators (e.g., expenses, beta ratios, click rates), designing structured workflows for reproducible data cleaning and transformation.
- Collaborated with cross-functional consulting teams to generate actionable recommendations, applying quantitative methods across economics and finance while supporting team-wide analytical consistency.

ENTREPRENEURSHIP EXPERIENCE

Co-Founder, *BigDog (A Decentralized Gamefi Simulation Platform with 100K USD funding)* **Mar. 2023 to Present**

- Designed and deployed blockchain-based experimental frameworks to simulate market microstructure strategies under uncertainty; attracted 20,000+ participants across 15+ countries, generating a clean dataset of over 1 million decision records for reproducible analysis.
- Built scalable Python pipelines for automated backtesting & simulation, processing 100,000+ simulated trades, reducing computational error variance by 25% and cut evaluation time by 40%, aligning with applied research best practices.
- Coordinated a distributed team of 10 developers and 8 analysts; mentored members in computational programming, agent-based modeling, and economic/financial theory, translating into 3 peer-review submissions in social science venues and adoption of models by 2 academic labs.

Co-Founder, *Howdream (Educational Art Therapy Company under the U35 List of China),* Canada **Jul.2022 to Dec. 2023**

- Implemented automation tools (UiPath, Git) to manage & document workflows for a volunteer research team of 30+ members, improving task throughput by 60%, reducing average task latency from 5 days to 2 days.
- Applied statistical model (SPSS) and time-series forecasting (LSTM) on engagement metrics from 1,200+ participants, improving predictive accuracy by 35%, which led to a 40% increase in program participation rate within 6 months.
- Facilitated interdisciplinary sessions integrating experimental design, econometrics, and predictive modeling; led to the development of 2 scalable evaluation tools adopted by partner institutions to assess learning & emotional outcomes post-engagement.

Co-Founder, *Vira (Online Educational Insitution,UN Member)* **Dec.2021 to Dec. 2023**

- Founded an online 3D virtual study room, including e-commerce portal, frontend, RESTful APIs, state management (Redux), database (MongoDB); enabled CRUD operations and supported daily concurrent users of 500+ across multiple time zones.
- Collaborated with 3 major hospitals to run comparative stress-testing studies using ANOVA & SEM (PyTorch & SPSS) on data from 1,200+ students, identifying strategies that boosted mental health workshop engagement by 40%, while reducing reported stress scores by 25% over a 8-week period.